

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460921.6562 +/- 0.0085 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.157 +/- 0.046
Transit depth (Rp/Rs)^2	2.45 +/- 1.44 [%]
Orbital Inclination (inc)	85.88 +/- 2.88
Airmass coefficient 1 (a1)	26.01 +/- 12.0
Airmass coefficient 2 (a2)	-2.97 +/- 0.73
Scatter in the residuals of the lightcurve fit is	47.3 %
Best Comparison Star	None
Optimal Aperture	4.18
Optimal Annulus	10.49
Transit Duration (day)	0.056 +/- 0.021

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.157 ± 0.046 vs 0.1326 (deviation 0.39σ)
Duration fit vs published	0.056 d vs 0.0737 d (deviation 0.62σ)
Mid-time O–C	-20.0 min
Depth SNR	2.5
Residual scatter	47.3 %

Light curve

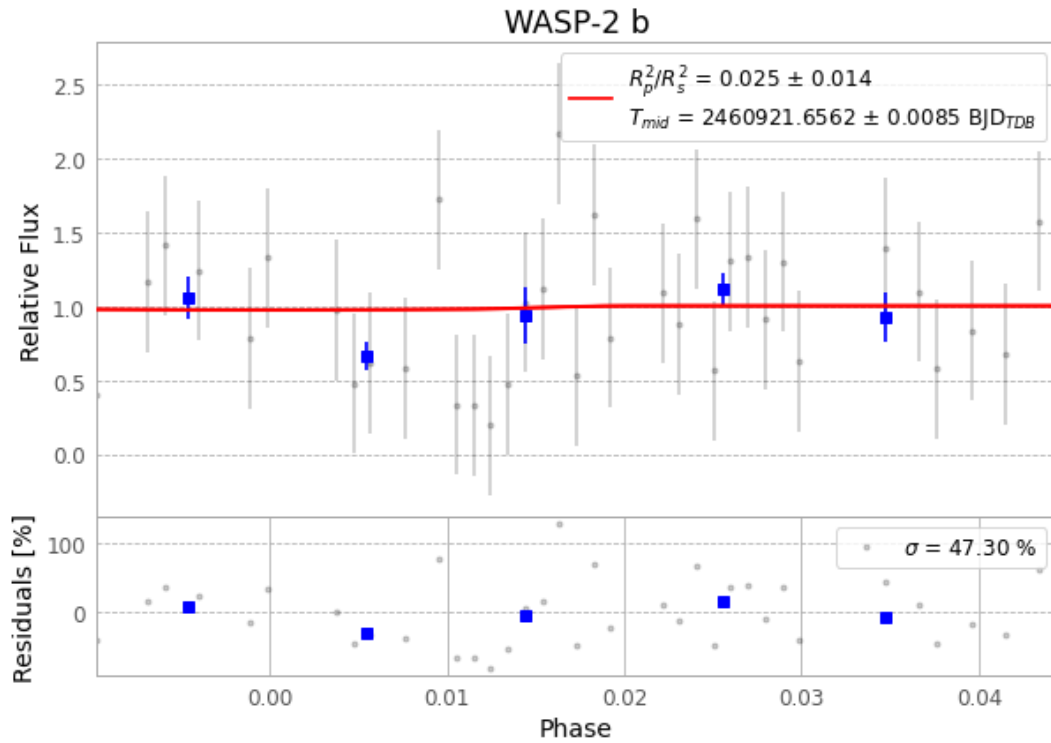


Figure 1 — FinalLightCurve_WASP-2 b_2025-09-02.png (SHA-256 f0cf4d5256dc1d5f...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [554, 440], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 8c6ee9df02bdd94b...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13

Data provenance

57 FITS frames (37 MB), WASP-2250903030616.FITS → WASP-2250903055415.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-2-250903.log (79f847417d1c...), FinalLightCurve_WASP-2 b_2025-09-02.png (f0cf4d5256dc...), FinalLightCurve_WASP-2 b_2025-09-02.csv (2c6d025b1134...)

Compute resources

Wall time 130.9 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-15 16:40Z.